

Reimagining Myntra with Generative AI

Myntra StyleGenie — an AI-native styling intelligence layer that transforms fashion discovery from passive browsing into personalised, intent-driven experiences.

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🔗 Live prototype: style-genie-myntra.vercel.app

01 The Problem: Fashion E-Commerce is Broken

Myntra serves **50M+ monthly active users** across India, offering **9 lakh+ products** from 5,000+ brands. Despite this scale, the platform suffers from fundamental UX inefficiencies that cost the business **over ₹2,000 Cr annually in returns alone**. The root cause: users navigate massive catalogs through keyword search and static filters — tools built for the desktop era, not for how people actually think about fashion.

70%

Cart Abandonment
Users add items but can't decide. Choice overload creates decision paralysis — the average user browses 40+ products before a single purchase.

35%

Return Rate
Fashion has the highest return rate in e-commerce. Primary driver: fit & style mismatch — shoppers can't try on, and size charts are unreliable.

3.2 min

Time to Find
Average time to find a relevant product. Users scroll pages of irrelevant results because search is keyword-based, not intent-aware.

₹2,100 Cr

Annual Return Cost
Reverse logistics, restocking, customer service & damaged goods. Pure cost — it generates zero revenue for the platform.

Who is affected? — User Segments

The Browser (62% of DAU)
Scrolls endlessly, rarely converts. Needs curated discovery — not more filters.

The Returner (31% return rate)
Buys 5, keeps 1. Fit & style mismatch is the core pain. Needs fit intelligence.

The Occasion Shopper (High AOV)
Needs a complete outfit for a wedding/party/fest. No idea where to start. Needs intent curation.

The Trend Chaser (18–28)
Wants celebrity/influencer looks at their budget. Needs visual search & trend matching.

Hypothesis: A Generative AI styling layer can reduce returns by 25%, lift conversion by 18%, and raise AOV by 22% — creating ₹1,200–1,700 Cr in annual impact.

02 Why Generative AI? Traditional Approaches Have Hit a Ceiling

Myntra already uses predictive ML (collaborative filtering, content-based recos). These work for simple cases but **fundamentally cannot** handle the problems above. Below: why each traditional approach fails, and how GenAI solves it — these are categorically different capabilities, not incremental improvements.

User Problem	Traditional ML	Why It Fails	GenAI Solution	How It Works
"Outfit for a college fest under ₹2K"	Keyword search + price filter	Can't understand occasion, vibe or styling context — only matches words	LLM conversational search	LLM parses intent (occasion + budget + style) → vector retrieval → composes outfits
"Find me something like what Alia wore"	Text-based image tags	Manual tagging can't capture aesthetic — "boho vibes" has no tag	CLIP multimodal visual search	CLIP encodes image + text in one space, finds visually similar catalog items
"Will this fit me?"	Static size-chart lookup	Ignores brand variation, fabric stretch, body diversity, user history	NLP review mining + body profile	Mines 10K+ reviews for fit signals, cross-refs the user's body data
"Style this kurta into a complete look"	"Frequently bought together"	Co-purchase ≠ styling. No colour, occasion or aesthetic sense	Generative outfit composition	LLM applies styling rules, generates occasion-specific outfits from real products
"What should I wear this week?"	"Trending" product list	Same list for everyone. No personalisation to individual style	AI trend digest + style graph	LLM generates trend narratives, curates to the user's evolving style profile

Key Insight — Traditional ML predicts what you might *click* next. GenAI understands what you actually *need*. It's the difference between a search engine and a personal stylist — categorically new capabilities enabled by LLMs, multimodal models and generative composition.

03 My Approach & the StyleGenie Solution

My approach: from problem to solution

01 DIAGNOSE

Studied 4 root pains: choice overload (40+ browses), fit uncertainty (35% returns), context blindness, manual discovery loop.

02 REFRAME

Shifted from inventory-first ("browse 9 lakh products") to intent-first ("look great at a wedding under ₹5K").

03 DESIGN

Mapped each segment + pain to one specific GenAI capability. Each feature solves exactly one root pain — no overlap.

04 VALIDATE

Anchored every feature to a measurable KPI (return rate, conversion, AOV, DAU). No feature ships without a metric.

The solution: 5 GenAI features (each mapped to a user pain)

StyleGenie Chat

AI Concierge

Solves: Choice overload

WHAT

Conversational AI that understands occasion, budget, body & style via natural language.

HOW

LLM + RAG: intent parse → vector search → outfit compose → stream.

VALUE

40 products → 5 outfits. 12% chat-to-cart.

Visual Match

Snap & Find

Solves: Style discovery

WHAT

Upload any photo (screenshot, Pinterest) → AI finds visually similar Myntra products.

HOW

CLIP encodes image → cosine similarity vs catalog embeddings → ranked results.

VALUE

35% CTR (vs 8% keyword search).

Outfit Studio

Composer

Solves: No outfit thinking

WHAT

Pick one product → AI generates 3–5 complete styled outfits using real catalog items.

HOW

Style-attribute encoding → LLM applies colour theory + occasion rules + composition.

VALUE

+1.8 items/order. 22% AOV lift.

FitGenius

Fit Predictor

Solves: Fit uncertainty

WHAT

Predicts your perfect size with 90%+ confidence using review mining + body profile.

HOW

NLP extracts fit signals from 10K reviews → cross-ref with user's body data.

VALUE

Returns 32% → <18%. ~₹525 Cr saved/yr.

TrendPulse

Trend Digest

Solves: Static discovery

WHAT

Personalised daily fashion feed with AI-curated trend lookbooks for your style.

HOW

Social-signal monitor → LLM trend narratives → products matched to trend + style graph.

VALUE

+40% DAU from daily engagement.

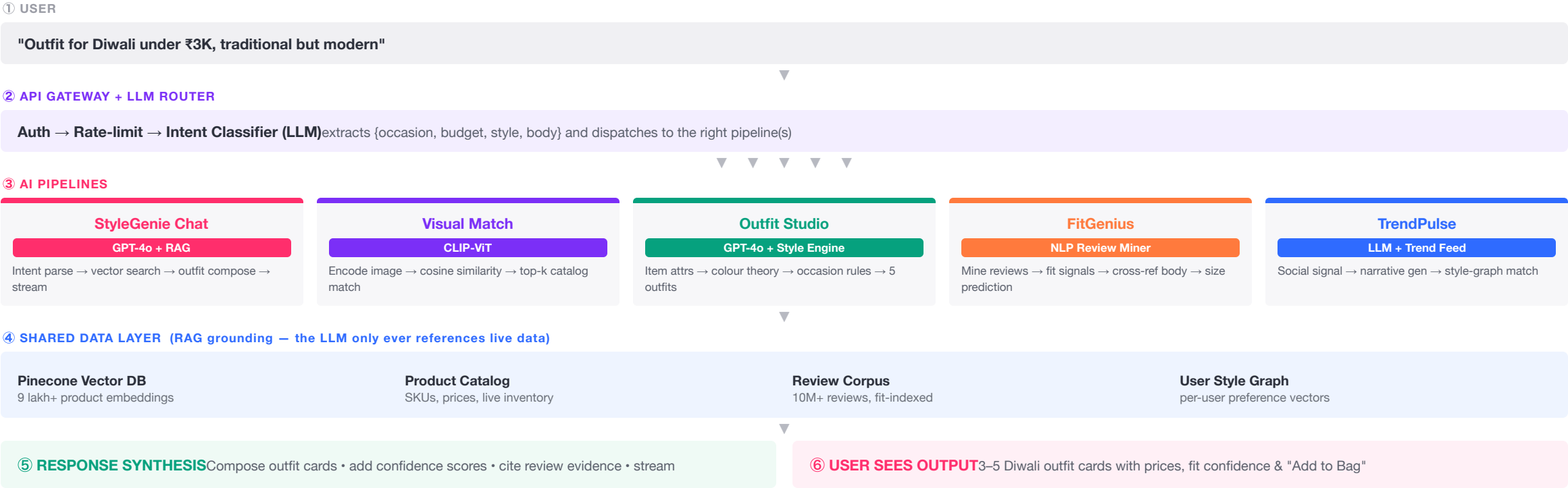
User segment → primary features: Browser → Chat + TrendPulse · Returner → FitGenius + Studio · Occasion → Chat + Studio · Trend Chaser → Visual + Pulse

Sources: CLIP — Radford et al. (OpenAI, 2021); RAG — Lewis et al. (Facebook AI, 2020); Myntra Category Report FY24.

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04 Implementation Flowchart: How StyleGenie Actually Works

End-to-end system flow: user query → AI orchestration → model layer → data retrieval → personalised response. Colour-coded by system layer.



05 The Live Prototype: How Each Feature Works In the App

A working React prototype rebuilt in Myntra's UI, with all five GenAI features interactive. Below is the **real in-app interaction** for each — what the user taps and what appears on screen — beside screenshots from the running app.

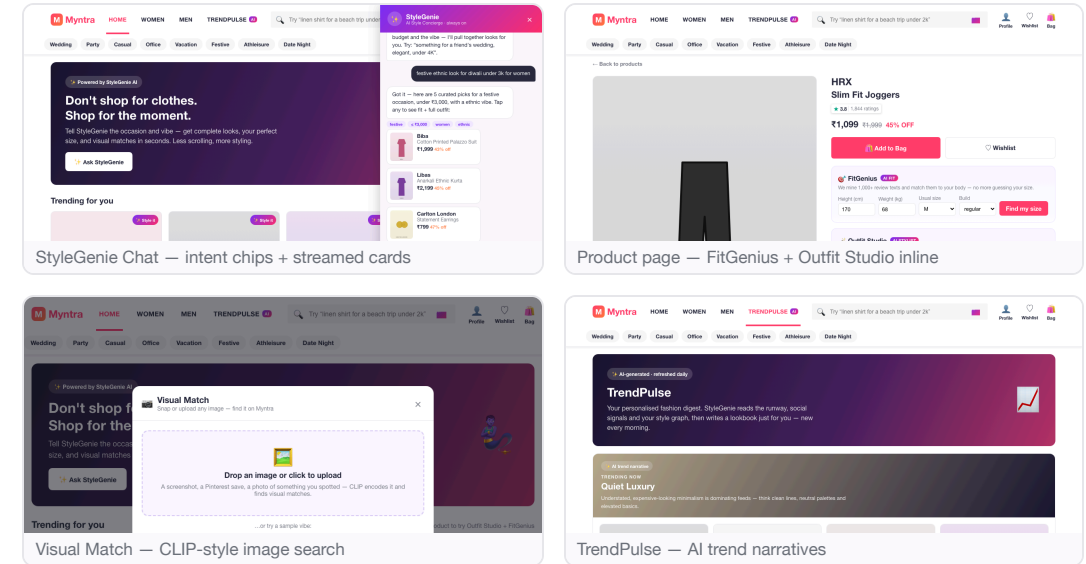
 **StyleGenie Chat** — tap the floating "🌟 Ask StyleGenie" button → type "festive look under 3k for women" → request appears as **intent chips** (festive · ≤₹3,000 · women) → **5 outfit cards stream in** → tap a card to open its product page.

 **Visual Match** — tap the  in the search bar → upload an image / pick a vibe → an "encoding with CLIP..." animation → a grid of products with % **match scores**.

 **Outfit Studio** — on any product page tap "Generate complete looks" → **3–5 styled looks** (top + bottom + footwear + accessory) each with a price total & occasion → "Add complete look to bag".

 **FitGenius** — enter height / weight / usual size / build → tap "Find my size" → size shown with a **confidence bar** + **review evidence** ("runs slim — we sized up").

 **TrendPulse** — open the **TrendPulse** tab → scroll **AI-written trend cards** (e.g. "Quiet Luxury"), each with a curated product row.



 **Live prototype:** <https://style-genie-myntra.vercel.app/> — append deep links `#genie/festive%20look%20under%203k`, `#p/12`, `#visual`, `#trends` to jump straight to a feature. **Interaction model:** a floating concierge + inline AI cards — never blocking the core flow. **Stack:** React + Vite; AI runs client-side for the demo, mapping 1:1 to the production backend in slide 04.

06 Success Metrics & Projected Business Impact

Each feature maps to a specific, measurable KPI tied to a business outcome. Projections use industry benchmarks for AI-driven personalisation in e-commerce.

25%

Return Reduction

FitGenius eliminates the #1 return driver (fit mismatch) via review mining + body data.

₹525 Cr saved/yr

18%

Conversion Uplift

StyleGenie cuts discovery from 40 items to 5 curated looks — shortening path to purchase.

Direct GMV lift

22%

AOV Increase

Outfit Studio drives bundle buying: the complete look instead of one item (+1.8 items/order).

+₹340/order

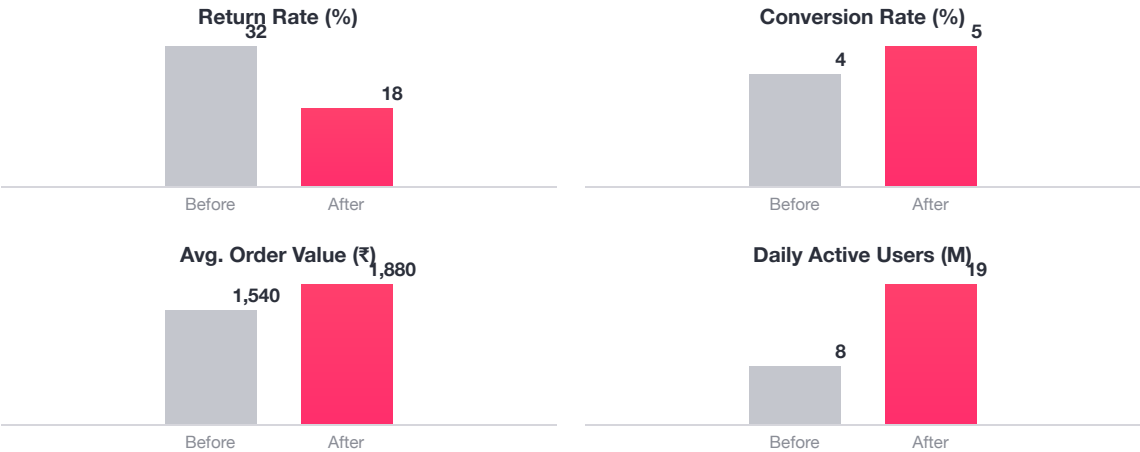
2.4X

Engagement

TrendPulse creates daily check-in behaviour — users return for AI-curated trends, not just to shop.

DAU/MAU ratio

Projected Impact: Before vs After



Feature-Level KPI Table

Feature	Metric	Target
StyleGenie Chat	Chat → cart	12%
Visual Match	Search CTR	35%
Outfit Studio	Items / order	+1.8
FitGenius	Return rate	<18%
TrendPulse	DAU lift	+40%

Total Projected Annual Impact: ₹1,200–1,700 Cr

Revenue uplift ₹800–1,200 Cr + Return savings ₹400–525 Cr + Premium subscription ₹50–80 Cr.

07 Risks, Pitfalls & Mitigation Strategy

Every AI product has failure modes. The key is not avoiding them — it's designing for them upfront. Below are the six most critical risks for StyleGenie, each with a specific, implementable mitigation.

 Hallucinations RISK AI suggests products that don't exist, are out of stock, or in the wrong category — destroying trust instantly. MITIGATION All responses grounded via a RAG pipeline — the LLM never invents product data, only references the live catalog. Confidence scoring filters low-certainty output; every card links to a real PDP. Critical	 Privacy RISK Users uncomfortable sharing body measurements, photos or style data. GDPR / India DPDP Act compliance. MITIGATION On-device image processing (CLIP runs locally). Explicit opt-in per data type. Zero third-party sharing. Full DPDP compliance with deletion on request. High	 Latency RISK LLM inference takes 3–5s — too slow for a shopping context where users expect instant results. MITIGATION Streaming responses (word-by-word reduces perceived wait). Pre-computed outfit caches. Edge-deployed CLIP. Target SLA <2s for all AI features. High
 User Trust RISK "How can a machine know what looks good on me?" Adoption depends on trust. MITIGATION Transparent confidence scores ("94% match"). Evidence-based explanations ("based on 10,247 reviews"). Manual override at every step. Trust earned through accuracy over time. Medium	 Scalability RISK GPT-4o costs at 50M+ MAU could reach ₹50–100 Cr/month — unsustainable without optimisation. MITIGATION Tiered models (Gemini Flash/Haiku for simple, premium only for complex styling). Aggressive caching — 100 similar queries share one response. Batch processing for TrendPulse. High	 Adoption RISK Users stick to browse-filter-scroll. AI features get ignored if presented as a separate, unfamiliar experience. MITIGATION Soft inline integration — AI appears within the existing flow, not a separate tab. Contextual nudges on struggle signals (long browse, repeated searches). Free tier to build habit, premium upsell later. Medium